

# ANKIT KUMAR

Delhi, India

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## EDUCATION

- **Birla Institute of Technology, Mesra** 2026  
*B.Tech in Electronics and Communication Engineering* CGPA: 8.29
- **Narbheram Hansraj English School** 2022  
*Indian School Certificate (ISC), New Delhi* Percentage: 95
- **Hill Top School** 2020  
*Indian Certificate of Secondary Education (ICSE), New Delhi* Percentage: 94.2

## EXPERIENCE

- **Samsung Research & Development, Noida** Jan 2026 - Jun 2026  
*SDE Intern* Noida
  - Developed a Flask-ADB diagnostic utility introducing illegal IMEI detection capability at the customer service tier, eliminating dependency on engineering escalation and reducing diagnosis time to under 60 seconds; currently in staged rollout targeting 300+ Samsung customer service centers nationwide.
  - Redesigned an internal file transfer server on Django and Nginx, replacing a WSGI-based architecture that hard-failed above 30 MB with a chunked multipart upload pipeline — client-side 10 MB packet segmentation, asynchronous server-side chunk reassembly, and SQL-backed chunk-state tracking — removing the file size cap, enabling resumable uploads with real-time progress tracking; deployed internally across 5 teams serving 200+ users.
  - Designed a Carrier Aggregation combo extraction engine that aggregated and indexed 500+ configuration entries across 10+ plain-text Qualcomm specification files into a unified searchable interface, reducing engineer lookup time by 95%+ (5+ minutes to under 10 seconds) and eliminating manual cross-referencing for the Communications Protocol team.
- **Tata Motors** May 2025 - Jul 2025  
*SDE Intern* Jamshedpur
  - Engineered a fault-isolated data pipeline replacing twice-weekly manual file transfers with 15-minute aggregation cycles across 11 assembly-line sensor systems, collecting 165+ Excel reports per run; architected per-source fault isolation to handle file locks, network drops, machine offline events, and concurrent write conflicts independently — ensuring zero-downtime pipeline continuity.
  - Introduced a real-time safety compliance system using YOLOv10 and Python, processing 5 simultaneous live CCTV feeds at 10 FPS on standard laptop hardware; introduced automated helmet violation detection with snapshot and bounding-box audit logging where no prior mechanism existed, achieving 94%+ detection accuracy across Bay 11 of the production line.

## PERSONAL PROJECTS

- **Inventory Management Microservices Backend**  Jun 2025  
*Tools & technologies : Spring Boot, MongoDB, OpenFeign, REST API, Kafka, Java MailSender, Docker*
  - Built a Spring Boot backend with 5 independent services (Product, Order, Supplier, Notification, Inventory) communicating via REST APIs using OpenFeign clients, with Eureka-based service discovery for dynamic inter-service routing and MongoDB for persistent storage.
  - Implemented Kafka-based async messaging between Order and Notification services and containerized the full system using Docker Compose for single-command deployment; used Spring Cloud Gateway as a unified entry point for all service endpoints.
- **QueryInsight : AI-Powered SQL Chatbot & Visualization Dashboard**  May 2025  
*Tools & technologies: Python, LangChain, Groq API, SQLAlchemy, MySQL, and Plotly/Dash.*
  - Engineered a multi-stage NL→SQL pipeline using LangChain (LLMChain) and Gemma2 9B via Groq API with dynamic schema injection via SQLAlchemy reflection — mapping table structures, column properties, and foreign-key relationships into zero-shot prompts — implementing a three-stage output layer (SQL extraction, syntax validation, schema cross-referencing) with strict read-only enforcement, eliminating invalid column/table hallucinations before execution.
  - Incorporates an auto-selecting visualization layer in Plotly/Dash that infers chart type from query result schema across 7 chart types — routing aggregations to bar charts, time-series to line charts, and multi-column results to comparative views — with anomaly detection and correlation analysis on returned datasets.
- **Predictive Maintenance (RUL Prediction)**  Feb 2025  
*Tools & technologies: Python, TensorFlow, XGBoost, Pandas, Random Forest, Scikit-learn*
  - Built a RUL prediction system on NASA C-MAPSS (FD001) by eliminating 7 redundant sensors via Pearson correlation, RUL clipping at 195 cycles, and 10-cycle rolling mean augmentation — achieving SVR regression performance of RMSE 32.48 (R<sup>2</sup>: 0.71) via 4-fold cross-validation, and extending to a 3-tier risk classifier (No Risk / Moderate / Critical) using SVC and Random Forest with stratified splits, reaching 68% validation accuracy across imbalanced classes.

## TECHNICAL SKILLS

**Languages:** Java, Python

**Backend & DevOps:** Spring Boot, REST APIs, Microservices, OpenFeign, Kafka, Docker, GitHub Actions (CI/CD)

**Databases:** MySQL, MongoDB

**AI / ML:** LangChain, Groq API, SQLAlchemy, Scikit-learn, XGBoost, SVR, Random Forest, YOLOv10, Plotly/Dash

**Tools & Concepts:** Git, ADB, TCP/IP, HTTP/HTTPS, OSI Model